

Robert Mirea

Software Engineer — Backend | Bucharest, Romania

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SUMMARY

Backend-focused software engineer building Python microservices on a high-scale, event-driven conversational AI platform. Hands-on with RabbitMQ, PostgreSQL/pgvector, Redis, Docker, and CI/CD, with production reliability ownership and comprehensive testing (unit, integration, end-to-end). Also experienced with Java/Spring Boot. Currently pursuing an MSc in Computer Science.

EXPERIENCE

Software Engineer — Tyrell Corporation (Remote) | Nov 2025 – Present

High-scale, real-time conversational AI platform (microservices, event-driven architecture).

- Build and maintain Python microservices in an event-driven architecture (RabbitMQ message broker, PostgreSQL/pgvector, Redis).
- Developed a generative image pipeline integrating ML inference services, with automatic failure recovery and credit/transaction integrity.
- Own production reliability: alerting (New Relic), incident response, and distributed-systems debugging across services.
- CI/CD, Docker, multi-service deployments; unit, integration, and end-to-end test suites.

Software Developer (Internship) — BearingPoint, Brasov | Jun 2024 – Aug 2024

- Built a Python application to manage and validate employee records, enforcing strict data formats and constraints.
- Used Pydantic for schema validation and PostgreSQL for storage; implemented duplicate prevention and structured exports for downstream analysis.

Application Developer (Internship) — Synergo Applications, Brasov | Aug 2023 – Sep 2023

- Designed and implemented a land registry management application in Java, applying OOP design principles to model ownership records and property transactions.

PROJECTS

Jarvis — Voice-Controlled AI Assistant — Python 3.12, FastAPI, TypeScript, Claude API, ElevenLabs | 2026

- Full voice loop (speech-to-text → Claude → text-to-speech) with turn-taking and barge-in, built on ElevenLabs Agents with Claude as the LLM.
- On-device client-tool executor: launches any installed app by voice using fuzzy name matching (Windows/WSL); FastAPI chat endpoint with multi-turn history, web UI, and pytest test suite. github.com/Robwert0/jarvis

Price Comparator — Java, Spring Boot | 2025

- Backend service comparing grocery prices across Romanian retailers (Lidl, Kaufland, Profi): daily optimized cart, discount tracking, price history, custom alerts, and unit-price (per kg/l) recommendations. github.com/Robwert0/Price_comparator

Real-Time Face Recognition Attendance System — Python, OpenCV, MediaPipe | 2024 – 2025

- Webcam-based attendance tracker: MediaPipe face-mesh detection, cosine-similarity embedding matching, Tkinter registration GUI, and duplicate-safe CSV logging — modular pipeline running in real time on standard laptops.

SKILLS

Languages	Java, Python, SQL, TypeScript, C++
Backend	Microservices, event-driven architecture, RabbitMQ, Redis, PostgreSQL/pgvector, Spring Boot, FastAPI, REST APIs, Pydantic
Ops & Testing	Docker, CI/CD, New Relic alerting, incident response; unit, integration, and end-to-end testing
Tools & Other	Git/GitHub, OpenCV, MediaPipe, scikit-learn, React, MATLAB/Simulink, ROS

EDUCATION

MSc, Automatic Control and Computer Science — National University of Science and Technology

Politehnica Bucharest | Oct 2025 – Present

BSc, Robotics — Faculty of Electrical Engineering and Computer Science — Transilvania University of

Brasov | 2021 – 2025

- Thesis-level projects in robotics: 6-DOF robotic arm — CAD design (CATIA), Simulink simulation, forward/inverse kinematics control.

LANGUAGES

Romanian (native) | English (C1)